

FRAUDSHIELD AI

Executive Fraud Risk Report

Dataset quality, fraud patterns, model evaluation, and investigation priorities

Version 1.0.0

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Dataset: sample_transactions.csv

DECISION-SUPPORT NOTICE

This report summarizes model signals for human review. A score is not proof of fraud and must not be the sole basis for blocking a transaction or penalizing a customer.

1. Executive Summary

15	11	97.1/100	1
TRANSACTIONS	FEATURES	DATA QUALITY	DUPLICATES

- Dataset quality is rated excellent at 97.1/100.
- Missing data affects 1.82% of all cells.
- 1 duplicate rows and 0 constant columns were detected.
- The current recommended model candidate is Logistic Regression, ranked by holdout PR-AUC.
- 8 transactions enter manual review; 4 are in the Critical band.

2. Data Quality

Measure	Result	Interpretation
Rows	15	Transactions available
Columns	11	Source fields available
Missing cells	3	1.82% of cells
Duplicate rows	1	6.67% of rows
Empty columns	0	Should be removed before training
Constant columns	0	No predictive variation
Formula-like cells	0	Review before spreadsheet export

3. Labeled Fraud Profile

Measure	Result
Target column	Class
Selected fraud label	1
Valid labeled rows	15
Fraud rows	4
Fraud rate	26.67%
Missing target rows	0
Fraud-linked amount	87,897.00
Fraud amount exposure	87.60%

4. Model Evaluation

All preprocessing and fitting occur inside leakage-safe pipelines. The leaderboard uses an untouched stratified holdout; PR-AUC is the primary ranking measure.

Model	Precision	Recall	F1	PR-AUC	ROC-AUC
Logistic Regression	0.800	0.750	0.774	0.820	0.910
Random Forest	0.760	0.700	0.729	0.790	0.890
Extra Trees	0.720	0.780	0.749	0.770	0.880

Training rows: 12 | Holdout rows: 3 | Holdout fraud rows: 1

5. Investigation Queue

Risk band	Transactions	Share
Low	4	26.67%
Medium	4	26.67%
High	3	20.00%
Critical	4	26.67%

Highest-risk queue entries

Row	Source index	Risk score	Band	Decision
9	9	95.00	Critical	Manual review
8	8	90.00	Critical	Manual review
14	14	85.00	Critical	Manual review
7	7	80.00	Critical	Manual review
6	6	70.00	High	Manual review
13	13	65.00	High	Manual review
5	5	60.00	High	Manual review
4	4	50.00	Medium	Manual review
3	3	40.00	Medium	No automatic hold
12	12	35.00	Medium	No automatic hold

Scoring model: Logistic Regression | Manual-review threshold: 0.50

6. Governance Recommendations

- Keep human review between model output and any customer-impacting action.
- Select thresholds using fraud loss, investigation capacity, and false-alert cost.
- Validate on future time periods when the production task predicts future transactions.
- Monitor data drift, fraud-rate drift, calibration, recall, and false-alert volume.
- Review proxy variables, access controls, retention rules, and audit logging before deployment.
- Retrain only through a versioned approval process with repeatable evaluation.

7. Methodology and Limitations

- Data-quality scoring measures completeness, uniqueness, and non-constant columns; it is not model accuracy.
- Correlation and feature influence do not establish real-world causation.
- Scores produced for training rows may be optimistic and must not replace holdout metrics.
- Small datasets can demonstrate workflow but cannot support reliable performance claims.
- Joblib artifacts must be loaded only from trusted sources because they use Python pickle.
- Production use requires authentication, encryption, privacy review, monitoring, and incident response.

8. Production Readiness Checklist

Control	Portfolio status	Production requirement
Authentication and authorization	Not included	Required
Encryption and secret management	Local only	Required
Decision and access audit logs	Basic app log	Required
Data retention and deletion policy	Not included	Required
Performance and drift monitoring	Not included	Required
Incident response process	Not included	Required
Human-review operating procedure	Conceptual	Required
Privacy, legal, and fairness approval	Not included	Required